

RETENTION AND RELAPSE

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LEARNING OBJECTIVES

- *Define retention and relapse*
- *Narrate different philosophies of retention*
- *Explain Causes of relapse*
- *Explain different Retention types*
- *Explain and clinically apply different types of retainers*

DEFINITION

- At sporting events, no matter how good things look for one team late in the game, the saying is “It’s not over until it’s over.”
- In orthodontics, although the patient may feel that treatment is complete when the appliances are removed, an important stage lies ahead.
- Any treatment is a failure unless the treatment results can be *retained*.
- Retention has been defined by Moyers as **“maintaining newly moved teeth in position, long enough to aid in stabilizing their correction”**
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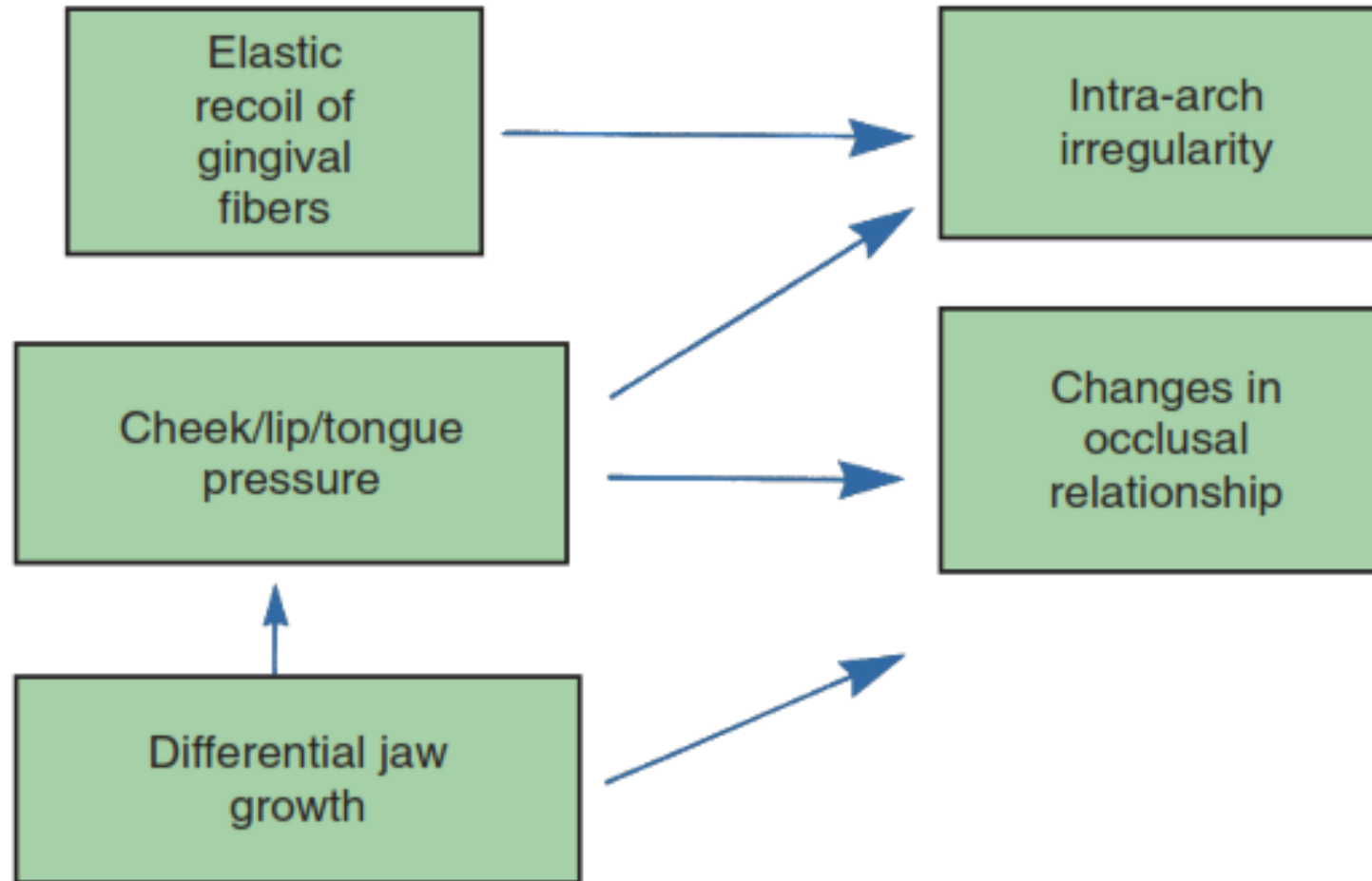
CONT'D

- **Relapse** has been defined as “**the loss of any correction achieved by orthodontic treatment**”
- The tendency of teeth to return back to the original position post-Orthodontic treatment
- **Stability** of the orthodontic outcome can only be achieved and maintained once the factors influencing relapse are identified and taken care of, right from the step of diagnosis, and continuing throughout the treatment

Why Is Retention Necessary?

- The gingival and periodontal tissues are affected by orthodontic tooth movement and require time for **reorganization** when the appliances are removed.
- The teeth may be in an inherently unstable position after the treatment, so **soft tissue pressures** constantly produce a relapse tendency.
- Changes produced by **growth** may alter the orthodontic treatment result.

➤ The major causes of relapse after orthodontic treatment:



Causes/Destabilizing Factors Attributing Relapse

1. Stretched fibers
2. Continued growth
3. Bone adaptation
4. Muscle imbalance
5. Persistent etiology
6. Erupting third molars
7. Unstable occlusion
8. Habits

- No single factors
- Relapse occurs due to a combination of causes

SCHOOLS OF THOUGHT / PHILOSOPHIES OF RETENTION

- *Various philosophies have been put forward over the years to explain post- treatment stability.*

1. THE OCCLUSION SCHOOL

- ❑ In 1880, Norman Kingsley
- ❑ suggested *that-"the occlusion of teeth is the most important factor in determining the stability in a new position.*
- ❑ Good interdigitation always aids in stability of treated results.

Cont...

2. THE APICAL BASE SCHOOL

- ❑ Alex Lundstrom, McCauley and Nance
- ❑ Alex Lundstrom: *the apical base is an important factor in the correction of malocclusion and maintenance of the stability of treated cases*
- ❑ McCauley: *the inter-canine and inter-molar widths should be maintained during orthodontic therapy to minimize retention problems*
- ❑ Nance: *the arch length cannot be permanently increased to a major extent*

Cont...

3. THE MANDIBULAR INCISOR SCHOOL

- The mandibular incisor school was proposed by Grieve and Tweed.
- They maintained that for reasons of stability, the mandibular incisors must be placed upright or slightly retroclined over the basal bone.

Cont

4.THE MUSCULATURE SCHOOL

- Paul Roger introduced the school of thought which included the necessity of establishing proper muscle balance.
- All the above philosophies are interrelated, e.g. retention depends on the occlusion established and the occlusion established must be within the bounds of normal muscle balance, also the occlusion and muscle balance established are dependent upon the amount of apical base available and the relationship of apical bases to one another.

RIEDEL'S THEOREMS OF RETENTION

- Riedel summarized all the different philosophies into nine theorems as follows.

THEOREM 1

"Teeth that have been moved tend to return to their former positions."

- Suggested influences include musculature imbalance, apical base, trans-septal fibers and bone morphology.

Cont

THEOREM 2

“Elimination of the cause of malocclusion will prevent relapse”

"The elimination of the causes of a malocclusion should aid in the retention of its correction." Approximately 25 percent malocclusions are due to a local cause, i.e. they are preventable.

- About 2 percent are congenital, e.g. cleft lip / palate and the other 70 percent are hereditary in origin based on racial admixtures and perhaps the gradual recession of the face beneath the cranium.
- All efforts should be made to correct the inherent etiologic factors in malocclusions, which are associated with an identifiable etiologic factor.

Cont...

- **THEOREM 3**

"Overcorrection of a malocclusion is a safety factor in retention," Overcorrection of deep bite, overbite and rotations is an accepted procedure.

- The rationale being that certain amount of relapse is to be expected after even the minutest of corrections.

Cont...

- **THEOREM 4**

"Occlusion is an important factor in retention." Proper interdigitation in post-treatment occlusion is important from the standpoint of stability.

- But even more important at times is the existence of reasonable occlusal balance during mastication and/or parafunction.
- The static occlusion should be examined, but importance should also be given to studying the functional occlusion (i.e., harmonious occlusal contacts during functional movements like mastication) of a case.

Cont....

- **THEOREM 5**

"Bone and adjacent tissues must be allowed to reorganize around of newly positioned teeth."

- The bone and the soft tissue surrounding the recently moved teeth require time to reorganize themselves.
- A reasonable period has to be provided before all restraining devices (retainers) can be done away with completely.
- The soft tissue envelope surrounding the oral cavity takes longer to orient itself to the new position of the teeth.

Cont...

- **THEOREM 6**

"Lower incisors must be placed upright over the basal bone".

More stable results are obtained when the mandibular incisors are either upright or slightly retroclined over the basal bone.

Cont...

- **THEOREM 7**
"Corrections carried out during period of growth are less likely to relapse."
- Early intervention is the norm of the day and has several advantages:
 - Prevents progressive, irreversible tissue damage.
 - Maximizes the use of growth.
- With concomitant tooth eruption, allows interception of malocclusion prior to excessive dental and morphologic compensations (which may become more difficult to correct and retain).
 - Allows corrections of skeletal malrelations while structures are morphologically immature and amenable to correction.
 - Prevents the development of psychological problems associated with malocclusions, as the cases are treated before the child becomes conscious of his/her appearance.

Cont...

- **THEOREM 8**

'The further teeth have been moved, the less the likelihood of relapse.'

- " Thus, when the teeth have been moved over greater distances, the retention required is decreased.

Positioning teeth far away from their original environment will produce equilibrium, is controversial.

Cont...

- **THEOREM 9**
"Arch form, particularly in the mandibular arch, cannot be permanently altered by appliance therapy." Altering the original arch form increases the risk of relapse.
- Studies have shown that:
 1. The mandibular inter-canine width cannot be increased, except in clearly defined situations.
 2. Premolar width can be increased approximately 2 mm and be confidentially expected to remain stable.
 3. *Molar* width cannot be increased, unless buccolingual axial inclinations are being corrected.
 4. Arch length will decrease throughout life, primarily before the age of 30 years.
- This generally manifests as uprighting of incisors, increasing the inter-incisal angle, crowding and displacement of anterior teeth and a recurrence of rotations.

THEOREM 10

- Moyers

“Many treated malocclusions require permanent retaining devices”

Timing of Retention

- Daily(essentially full-time)for the first 3 to 4 months
- Continued on apart-time basis for at least 12 months to allow time for remodeling of gingival tissues.
- If significant growth remains,continued part-time until completion of growth.

Types Of Retention :Retention planning

- Reidel has grouped retention planning into 3 groups

1. Natural/No retention required

A. Cross bites

✓ *Anterior*: When adequate overbite has been achieved

✓ *Posterior*: When axial inclination of teeth remain reasonable after correction

B. Dentitions treated with serial extractions

C. Corrections achieved by retardation of maxillary growth once the patient has completed growth

D. Block-out or highly place canines in class I extraction cases

Cont...

2. Limited Or Short-term Retention

- A. Class I non-extraction cases with spacing and protrusion of maxillary incisors (until normal lip and tongue function has been achieved)
- B. Class I and II extraction cases
- C. Early correction of rotated teeth to their normal position before root completion
- D. Cases involving ectopic eruption or the presence of supernumerary teeth
- E. Corrected deep bites

Cont...

3. Prolonged Or Permanent Retention

A. In many cases, to maintain existing esthetics extraction may not be done.

The only way to create space in such cases is through **expansion**.

These cases, especially in mandibular arch require permanent or semi-permanent retention.

B. Cases of considerable severe generalized spacing

C. Severe rotation or severe labiolingual malposition

D. Spacing between maxillary central incisors with an otherwise normal occlusion

E. Expanded arches in CLP patients

TYPES OF RETAINERS

- *Retainers are passive orthodontic appliances that help in maintaining and stabilizing the position of a single tooth or group of teeth to permit reorganization of the supporting structures.*
- Retainers can be simply classified as *removable* and *fixed*, based on the patient's ability to remove the appliance.

IDEAL REQUIREMENTS OF RETAINING APPLIANCES

Graber put forward certain criteria that any retaining appliance should possess. These include;

1. It should restrain each tooth in its direction of relapse
2. It should permit the forces associated with functional activity to act freely on the teeth, permitting them to respond in as nearly a physiologic manner as possible.
3. It should be as self-cleansing as possible and should be reasonably easy to maintain optimal hygiene.
4. Should be as inconspicuous as possible, esthetically good.
5. Strong enough to bear the rigors of day-to-day usage.

❑ Factors which should be considered in choosing of the type of retainer:

- 1. Type of the malocclusion treated*
- 2. The aesthetic needs(concerns)*
- 3. Oral hygiene of the patient*
- 4. Patient cooperation*
- 5. The duration of retention phase*
- 6. Cost of the retainer*
- 7. Preference of the patient*
- 8. Cases which need habit trainer or breaker*

REMOVABLE RETAINERS

1. Hawley's retainer and its modifications
2. Begg wrap-around retainer (with long labial bow)
3. Clip-on retainer (spring aligner)
4. Kesling tooth positioner
5. Invisible retainers (Essix retainer)

FIXED RETAINERS

1. Passive fixed orthodontic appliance
2. Banded canine to canine lingual retainers
3. Bonded lingual retainer (with modifications)
4. Band and spur retainer

REMOVABLE RETAINERS

These are the most frequently delivered retainers in routine clinical practice.

Removable retainers are passive appliances that can be removed and reinserted by the patient.

1. Hawley's retainer(STANDARD): Charles Hawley designed the Hawley's retainer in 1920

It is the most popular removable appliance in use today.

The retainer consists of a labial bow extending from canine to canine with retentive clasps (Adam's) on the first permanent molars.

It has good retention and is capable of preventing the anterior teeth from rotating or developing gaps.

The design by itself prevents the extraction spaces from opening up and is capable of closing minor spacing in the anterior segment.

The clasp locations for a Hawley retainer must be selected carefully because clasp wires crossing the occlusal table can disrupt rather than retain the tooth relationships established during treatment. **Circumferential clasp** so the terminal molar may be preferred over the more effective Adams clasp if the **occlusion is tight**.



Cont....

Hawley's retainer with long labial bow:

A simple modification to the original appliance where the labial bow has 'U' loops on the premolar distal to the canine .

- This modification allows the closure of spaces distal to the canine.



Cont....

- ***Hawley's retainer with contoured labial bow:***

Again a modification of the original Hawley's retainer, the labial bow is contoured to the anterior teeth.

The advantage is of better control over the anterior teeth.



Fig. 23.8B: Retainer with long labial bow soldered to Adam's clasp

Cont....

- ***Hawley's retainer with continuous labial bow soldered to clasps .***

This allows for space closure in the anterior as well as the extraction/premolar region.

It is well tolerated by the patient and prevents spaces opening up in the region where the extractions were undertaken.



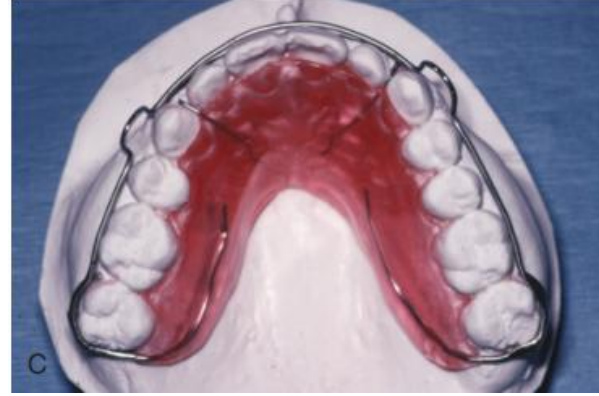
Fig. 23.10: Retainer with labial bow soldered to Adam's clasp

2. Clip-on Retainer/ spring aligner/

- Can be both active or passive retainer
- Has both labial and lingual wire component embedded with in the acrylic
- Can be used to bring about correction of rotations commonly seen in lower anterior



3. Wrap Around Retainer



- Extended version of the spring aligner that covers all the teeth
- Consists of wire that passes along the labial and lingual surfaces of all erupted teeth which is embedded in a strip of acrylic
- Indicated in stabilizing a periodontally weak dentition



4. Clear (Vacuum-Formed) Retainers

- A retainer made with a clear heat-softened plastic that is sucked down tightly over the teeth with a vacuum-forming device is another form of the older wraparound retainer made with acrylic and wire.
- Because the material is transparent and thin, a vacuum-formed retainer is all but invisible, and most patients prefer them.
- At present this is the most widely used retainer for the maxillary arch, and patients using a clear retainer report greater satisfaction with their treatment than those with other types of retainers.



➤ limitations to vacuum-formed retainers:

- The thickness of the material over the occlusal surface of the teeth can become a problem, especially if both the upper and lower arches are retained in this way, because separation of the posterior teeth in occlusion may develop.
- The combination of a vacuum-formed upper and a fixed lower retainer is better than maxillary and mandibular vacuum-formed retainers.
- The retainer maintains alignment but does not control deepening of the bite as well as a palate-covering Hawley retainer.
- After 6 to 9 months, a vacuum-formed retainer tends to crack and discolor to the point that it has to be replaced.

5. Kesling's tooth positioner

- Kesling developed the tooth positioner in 1945.
- It is made up of thermoplastic rubber material and covers the upper and lower clinical crowns and part of the adjacent gingiva.
- The patient is unable to speak with the appliance in place.
- Minor adjustments for settling of occlusion can be made with this appliance.
- A tooth positioner also can be used as a removable retainer, either fabricated for this purpose alone, or more commonly, continued as a retainer after serving initially as a finishing device.
- A positioner does have one major advantage over a standard removable or wraparound retainer, though; it maintains the occlusal relationships as well as intra-arch tooth positions.
- A recent review indicated that positioners of this type were as effective as the combination of a maxillary vacuum-formed retainer and a bonded lingual incisor retainer with or without interproximal stripping.
- For a patient with a tendency toward class iii relapse, a positioner made with the jaws rotated somewhat downward and backward may be useful.



Fig. 23.15: Kesling's tooth positioner



FIXED RETAINERS

- Fixed retainers are generally cemented or bonded to the teeth.
- They are generally used in situations where intra-arch instability is anticipated and prolonged retention is planned .
- They are gaining popularity due to their relative invisibility and reduced dependence on the patient for wearing .

Indications for fixed retainers

1. Maintenance of lower incisor position during late mandibular growth
2. Following closure of diastemas
3. Maintenance of bridge pontic space
4. Compromised periodontal conditions with the potential for post-orthodontic teeth migration
5. Prevention of rotational relapse
6. Prevention of relapse after the correction of palatally erupted canines
; especially if proper overbite has not been achieved
7. Prevention of opening up of closed extraction space, especially in adult patients

Advantages of fixed retainers

1. Reduced need for patient cooperation
2. Can be used when conventional retainers cannot provide same degree of stability
3. Bonded retainers are more esthetic
4. There is no tissue irritation unlike what may be seen in tissue bearing areas of Hawley's retainer
5. There is no jiggling of teeth, which are present when removable appliances are removed and reinserted
6. Recall visits are reduced
7. Can be used for permanent and semipermanent retention
8. Are better tolerated by the patient.
9. Do not affect speech, which is frequently affected when removable retainers with a base plate are worn

Disadvantages of fixed retainers

1. More cumbersome to insert
2. Increased chair-side time
3. More expensive
4. Banded variety may interfere with oral hygiene maintenance
5. Are more prone to breakages as compared to the removable retainers
6. Loss of healthy tooth material
7. Tend to discolor
8. Potential sights of sensitivity and caries
- 9 Maintenance of oral hygiene might get compromised
10. If fracture, replacement is tedious

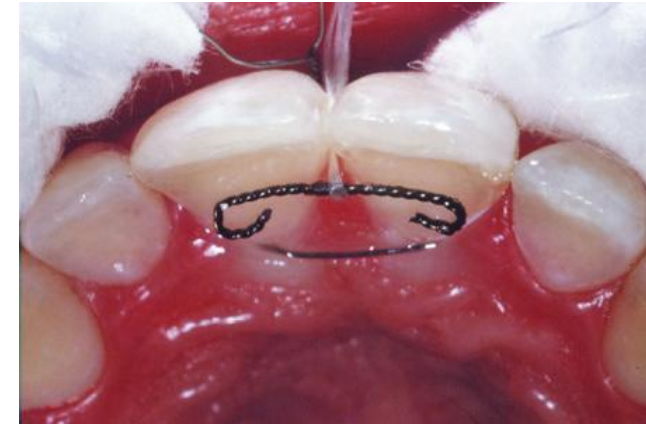


b/ fixed lingual retainer:

- flexible wire, braided steel archwire of 17.5-mil diameter



c/A section of twist wire



a/ Fixed lingual bar

- 28- or 30-mil steel (heavy enough)

d/ heavy intracoronal wire bonded to the adjacent teeth

